PH7002: Concepts in Statistical Mechanics
§1 Introduction

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Aperitifs

Reading

Sethna chapters 3 and 6 and Yeomans chapters 1 and 2.

In this chapter, we will quickly review the useful concepts in statistical mechanics used throughout this course. Assuming that this is familiar material, the main aim will be to gather together the relevant formulae in a form suitable for reference.

Important concepts

1. Microcanonical Ensemble
2. Canonical Ensemble
3. Thermodynamics
4. Free energies and Legendre transformation

Ludwig Eduard Boltzmann

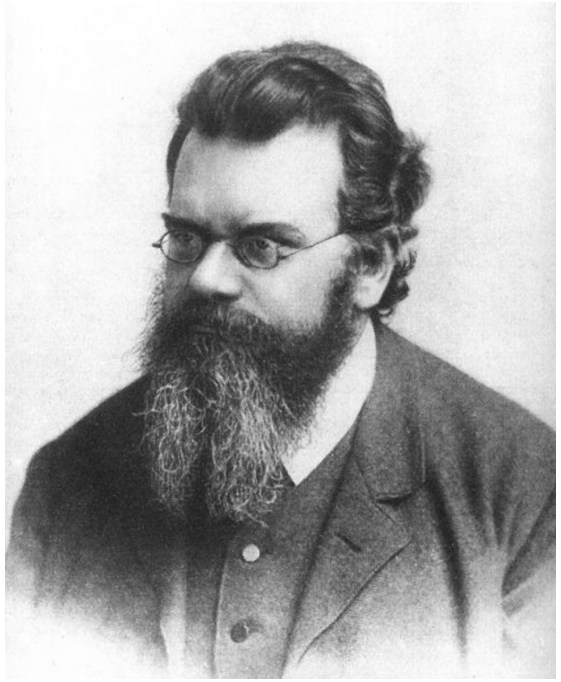


Figure: **Ludwig Eduard Boltzmann** (20 February 1844 – 5 September 1906) was an Austrian physicist and philosopher. His greatest achievements were the development of statistical mechanics, and the statistical explanation of the second law of thermodynamics. In 1877 he provided the current definition of **entropy**, $S = k_{\mathrm{B}} \ln \Omega$, interpreted as a measure of statistical disorder of a system.

Formulation

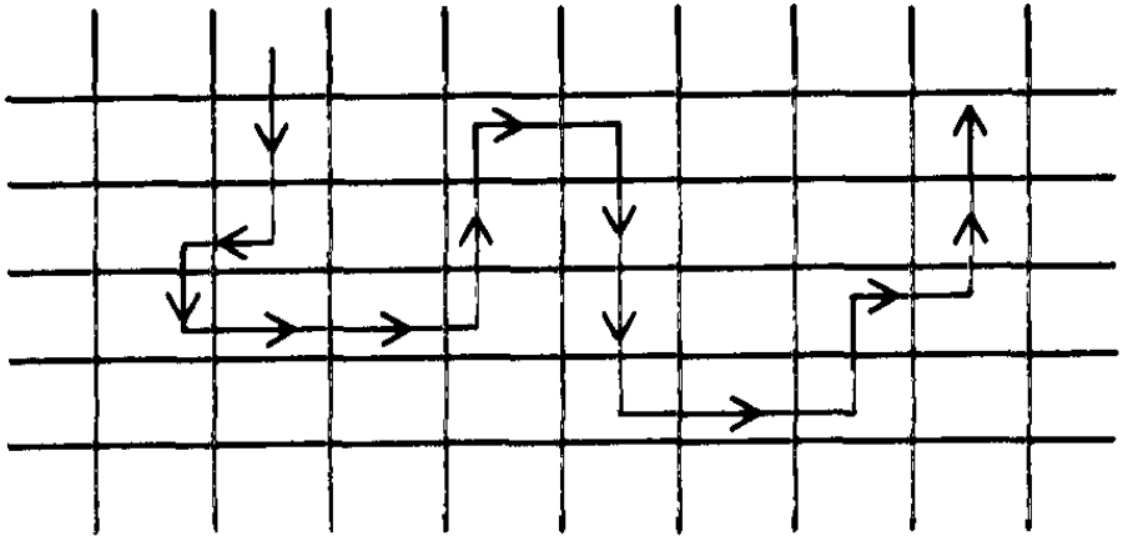
Consider the general equilibrium behavior of N atoms in a box of volume V —any kinds of atoms, in an arbitrary external potential. How can we solve for the behavior of our atoms? We can in principle determine the positions and momenta of the particles at any future time given their initial positions and momenta using Newton's law.

Some terminologies

- **System:** part of the universe we are interested in, only weakly coupled to the rest of the universe so that the dynamics is dominated by the internal interactions.
- **Isolated:** idealization — eliminate all external influences, in particular no forces, energy flux, particle flux etc. so that quantities conserved by the internal dynamics are strictly constant for the isolated system.
- **Thermodynamic Equilibrium:** wait long enough so that all transient behavior dies out, and measurements of macroscopic quantities become time independent.
- **Microstate:** a complete microscopic specification of the particles so that the dynamics of the whole system could be followed in principle. Basically where particles are, and their velocities, for a classical system, and the quantum state (wave function) for a quantum system.

Phase space trajectory

Now try to think about this time evolution–the trajectory–of a many-body system. Microstates are defined by points in phase space.



For example, we might fix the total energy, E, the total number of particles, N, and the volume, V. These constraints cause the trajectory to move on a "surface" of state space-though the dimensionality of the surface is still enormously high.

Ensemble Average

A basic concept in statistical mechanics is that if we wait long enough, the system will eventually flow through (or arbitrarily close to) all the microscopic states consistent with the constraints we have imposed to control the system. Suppose this is the case, and imagine that the system is constantly flowing through the state space as we perform a multitude N of independent measurements on the system. The **observed value** ascertained from these measurements for some property G is

where G_i is the value during the i-th measurement whose time duration is very short-so short, in fact, that during the i-th measurement the system can be considered to be in only one microscopic state.

Then we can partition the sum as

where $G_\nu = \langle \nu | G | \nu \rangle$ is the **expectation value** for G when the system is in state ν. The term in square brackets is the **probability** or weight for finding the system during the course of the measurements in state ν.

Remember, we believe that after a long enough time, all states are visited. We give the probability or fraction of time spent in state ν the symbol P_ν and write

The averaging operation, i.e., the weighted summation over G_ν indicated by the pointed brackets, $\langle G \rangle$, is called an **ensemble average**.

Concept of Ensemble

An **ensemble** is the assembly of all possible microstates–all states consistent with the constraints with which we characterize the system macroscopically. For example, the **microcanonical ensemble** is the assembly of all states with fixed total energy E, and fixed size (usually specified by number of molecules, N, and volume, V). The **canonical ensemble**, another example, considers all states with fixed size, but the energy can fluctuate. The former is appropriate to a closed isolated system; the latter is appropriate for a closed system in contact with a heat bath.

Principle of Ergodicity

The idea that we observe the ensemble average, $\langle G \rangle$, arises from the view in which measurements are performed over a long time, and that due to the flow of the system through state space, the **time average is the same as the ensemble average**. The **equivalence** of a **time average** and an **ensemble average**, while sounding reasonable, is not at all trivial. Dynamical systems that obey this equivalence are said to be **ergodic**. It is difficult, in general, to establish the principle of ergodicity, though we believe it holds for all many-body systems encountered in nature.

Microcanonical Ensemble

The basic idea of statistical mechanics is, therefore, that during a measurement, every microscopic state or fluctuation that is possible does in fact occur, and observed properties are actually the averages from **all the microscopic states**. To quantify this idea, we need to know something about the probability or distribution of the various microscopic states. This information is obtained from an assumption about the behavior of many-body systems, which brings us to Boltzmann's postulate of equal a priori probabilities.

Boltzmann's postulate

Boltzmann's postulate of equal a priori probabilities

For an isolated system with fixed total energy E, and fixed size (perhaps speciied by the volume V and numbers of particles $N_1, N_2, \dots$) all microscopic states are **equally likely** at **thermodynamic equilibrium**.

In the microcanonical ensemble, all points in phase space (with a given energy) are a priori equally likely, so the average should treat them all with equal weight.

In other words, the macroscopic equilibrium state corresponds to the most random situation–the distribution of microscopic states with the same energy and system size is entirely uniform. According to the statistical assumption, the probability of a macroscopic state ν for an equilibrium system is

for all states in the ensemble. For states outside the ensemble, for example those for which $E_\nu \neq E$, P_ν is zero. This ensemble, which is appropiate to a system with fixed energy, volume, and particle number–the assembly of all microstates with these constraints–is called a **microcanonical ensemble**.

A dynamical system evolves along its energy surface, that is, the set of configurations in phase space $\{q_i, p_i\}$ with fixed energy $H(q_i, p_i) = E$. On this surface, there are special points with simple interpretation, for example, for an ideal gas, the points where all of the atoms are in the same small volume with almost identical energies ("atypical" microstate). But, very quickly, this point will evolve to a more generic point at which the atoms are scattered through the volume and have a distribution of momenta ("typical" microstate).

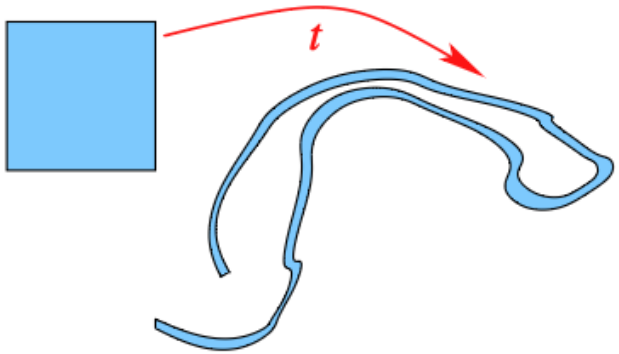
Atypical Typical

Boltzmann's insight is that, first, these generic points are macroscopically indistinguishable, and, second, that any one (that is, almost any point on the energy surface) has the macroscopic properties of the state of thermal equilibrium. In the small, these states are constantly changing into one another, but, in the large, there is no visible change. The postulate implies that we can calculate the macroscopic properties of any large assemblage of atoms as those of any **typical** state of given energy.

Liouville's theorem

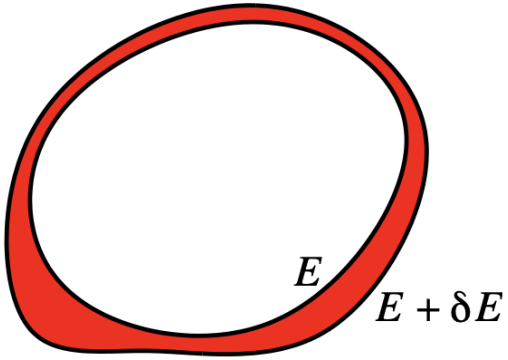
Why do we integrate over states using a uniform density in phase space rather than some other measure?

There is a well-defined answer to this question. In classical mechanics, there is a theorem, **Liouville's theorem**, that states that volumes in phase space are not changed under Hamiltonian time evolution. One can show that the volume in phase space, $d\mathbb{P}d\mathbb{Q}$, is preserved when we integrate the equations of motion. Thus, the microcanonical average as defined is **time-independent**, as would be appropriate for a description of equilibrium.



We calculate the properties of our ensemble by averaging over states with energies in a shell $(E, E + \delta E)$ taking the limit $\delta E \to 0$. Let us define the function $\Omega(E)$ to be the phase-space volume of this laws:

Here $H(P, Q)$ is the Hamiltonian for our system.



Finding the average $\langle O \rangle$ of a property in the microcanonical ensemble is done by averaging $O(P, Q)$ over this same energy shell:

More formally, one can write the energy shell $E < H(P, Q) < E + \delta E$ in terms of the Heaviside step function $\theta(x)$, where $\theta(x) = 1$ for $x \geq 0$, and $\theta(x) = 0$ for $x < 0$. We see that $\theta(E + \delta E - H) - \theta(E - H)$ is one precisely inside the energy shell. In the limit $\delta E \to 0$, we can write $\Omega(E)$ as a derivative:

Also, note that the derivative of the Heaviside function is the Dirac δ-function. Thus we have

Thus the microcanonical ensemble can be written as a probability density

or

in phase space. The expectation of a general operator O as

We can also consider as a definition of entropy the quantity

where $k_B = 1.381 \times 10^{-23}$ m^2 kg s^{-2} K^{-1} is the Boltzmann's constant. Notice that S defined in this way is **extensive** since if the total system were composed of two independent subsystems, A and B, with number of states andΩ_A and Ω_B separately, then the total number would be $\Omega_A \Omega_B$. That is,

The temperature T is determined from the derivative

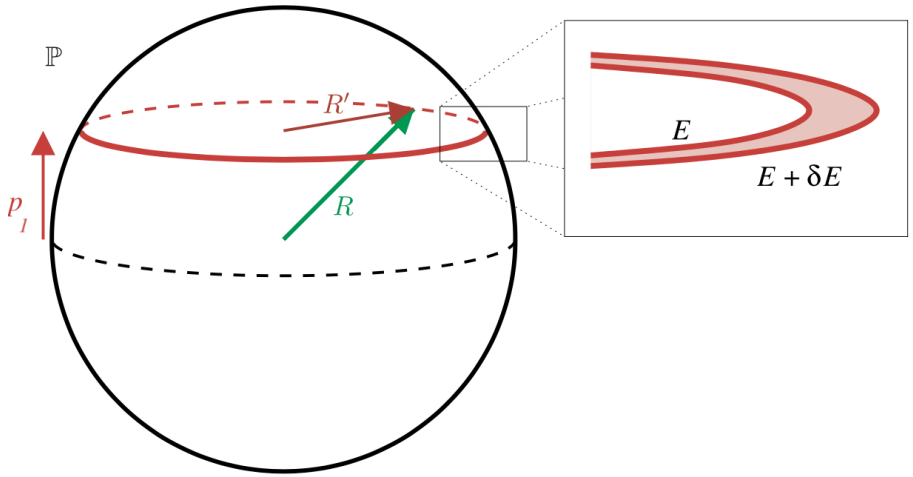
Therefore,

The thermodynamic condition that temperature is positive requires that $\Omega(N, V, E)$ be a monotonic increasing function of E. For macroscopic systems encountered in nature, this will always be the case.

Microcanonical ideal gas

The kinetic energy for particles (whether or not they interact) is

where the last form assumes all of our atoms have the same mass m. Hence the condition that a system of equal-mass particles has energy E is that the system lies on a sphere in $3N$-dimensional momentum space of radius $R = \sqrt{2mE}$.



To compute averages, we need to compute $\Omega(E)$ which is proportional to the area of the hypersphere. There's a trick to work out the area of a unit sphere in d-dimensions. Consider

In polar coordinates, this is

where A_d is the area of the unit sphere in d-dim. Change variables $y = x^2$ and $dy = 2xdx$

The integral is the Gamma function

Basic property of $\Gamma(z)$ is the recursion formula

derived by integration by parts

It is easy to see that

For $n =$ integer,

Also,

Thus,

For our problem, $\Omega(E)$ requires the area of a sphere of radius $P = \sqrt{2mE}$ and thickness

Then we have

What is the probability that a given atom has momentum $\mathbf{p}$? We can express this probability as a distribution.

To compute $f(\mathbf{p})$, recognize that if one particle has energy $p^2/2m$, momentum $\mathbf{p}$, then the remaining $N-1$ particles must have energy $E - p^2/2m$. Then,

Let us work out the dependence on p.

Typically, the energy is divided more or less equally among all particles, so

Using

The factor $2E/3N$ will appear alot. It is in fact the "temperature"

Then we find

This distribution of momenta is the famous Boltzmann distribution. Let us check the normalization.

Apply Stirling's approximation ($\ln n! \approx n \ln n - n + \frac{1}{2} \ln(2\pi n)$):

We can easily find that

In fact, we find that

This is the equipartition theorem.

Since the number of microstates is large, it is useful to define

Two systems in contact

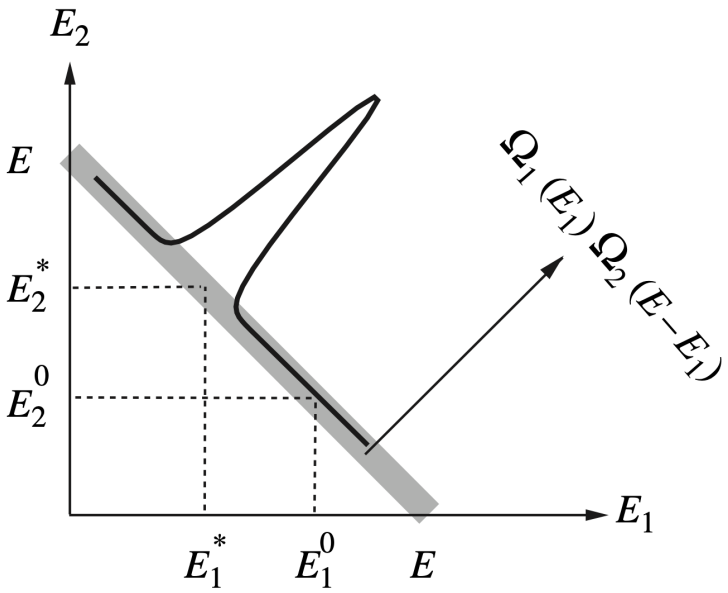
Given 2 assemblage of particles, what happen when they are brought into contact? Which way does energy flow? What state of each system results from this process?

1. Each initial state represented by a microcanonical ensemble of fixed initial energy.
2. As they exchange energy, they evolve on the energy surface of the total system.
3. Eventually, they reach a typical state of the system.

Mathematically, we have

Use the saddle point approximation

At equilibrium, we find that



Although all joint microstates are equally likely, the above results indicate that there is an exponentially larger number of states in the vicinity of (E_1^*, E_2^*). Originally, the joint system starts in the vicinity of some point (E_1^0, E_2^0). After the exchange of energy takes place, the combined system explores a whole set of new microstates. The probabilistic arguments provide no information on the dynamics of evolution amongst these microstates, or on the amount of time needed to establish equilibrium. However, once sufficient time has elapsed so that the assumption of equal a priori probabilities is again valid, the system is overwhelmingly likely to be at a state with internal energies (E_1^*, E_2^*). At this equilibrium point, the two systems are at the **same temperature**.

Example 1.1: Negative Temperature

Sethna 6.3(a), (b) - Negative Temperature.

The energy of the system for a given state is

where we have chosen the ground state as the zero of energy. The degeneracy of the mth energy level is the number of ways one may choose m objects from a total of N. That is,

The entropy and temperature are given by

Using Stirling's approximation, we find that

Combining this result with the formula for β yields

As a result, the energy $m = E\epsilon + N/2$ as a function of temperature is

which is 0 at $T = 0$ (i.e., only the ground state is populated), and it is $N\epsilon/2$ as $T \to \infty$ (i.e., all states are then equally likely).

Note that this would be more tricky without approximations because we would have to take a finite difference of $m!$ instead of a derivative of m. For $E > 0$, the argument of the logarithm is less than one, making the temperature $T < 0$!

Numbers

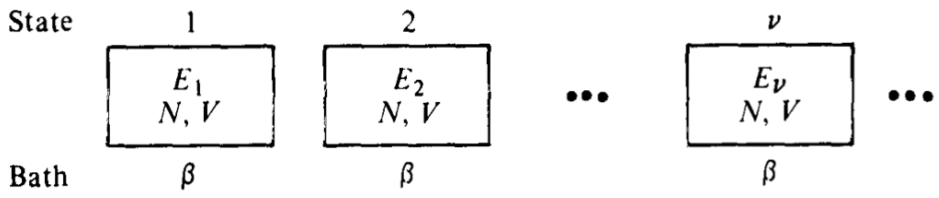
Definition

1. Intenstive quantities are quantities that remain constant as $N \to \infty$.
2. Extensive quantities are quantities that grow proportional to N as $N \to \infty$.
3. Super-extensive quantities are quantities that grow as to e^{cN} as $N \to \infty$.

The quantity $\Omega(E)$ is super-extensive. Hence it makes sense to define $S(E) = k_B \log \Omega(E)$. Then $S(E)$ is extensive as required.

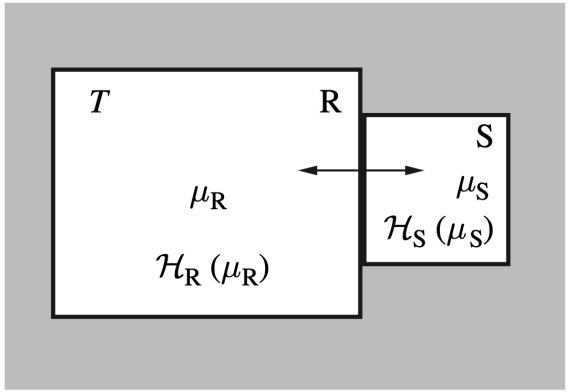
Canonical Ensemble

We consider now the **canonical ensemble** - the assembly of all microstates with fixed N and V. The energy can fluctuate, however, and the system is kept at equilibrium by being in contact with a heat bath at temperature T (or inverse temperature β). Schematically, we might picture the ensemble as shown.



The states we refer to here with the label ν are states of definite energy-eigenfunctions of Schrodinger's equation, $\mathcal{H}\psi_\nu = E_\nu \psi_\nu$.

To begin, consider the case where the bath is so large that the energy of the reservoir, E_R, is overwhelmingly larger than the energy of the system, E_S. Further, the reservoir is so large that the energy levels of the reservoir are a continuum and $d\Omega/dE$ is well defined. The energy in the system fluctuates because the system is in contact with the bath, but the sum $E = E_R + E_S$ is a constant.



Therefore, according to the statistical postulate–the principle of equal weights– the equilibrium probability for observing the system in state ν obeys

Since $E_S \ll E$, we can expand $\log \Omega(E - E_S)$ in the Taylor series

We choose to expand $\log \Omega(E)$ rather than $\Omega(E)$ itself because the latter is a much more rapidly varying function of E than the former.

By retaining only those terms exhibited explicitly in the expansion (which is valid because the bath is considered to be an infinite thermal reservoir), and using the statistical definition of temperature, we obtain

which is the canonical (or Boltzmann) distribution law. The constant of proportionality is independent of the specific state of the system and is determined by the normalization requirement

Hence,

The normalization

is known as the (canonical) **partition function**, and $\beta = 1/k_B T$, where "Tr" denotes the trace of a matrix (in this case, the trace of the Boltzmann operator matrix). It is a remarkable property of traces that they are independent of the representation of a matrix. This follows from: $\mathrm{Tr}\ A = \mathrm{Tr}\ SS^{-1}A = \mathrm{Tr}\ S^{-1}AS$. Thus, once we know $\mathcal{H}$ we can use any complete set of wavefunctions to compute Z.

The average of any observable A can be calculated as

As an instructive example of its use, consider the calculation of the internal energy $E(P, N, V)$, which is $\langle E \rangle$ in the canonical distribution:

which suggests that $\ln Z$ is a familiar thermodynamic function. In fact, the **Helmholtz free energy** is defined as

Often, our aim will be to calculate the free energy, from which we can calculate the relevant physical quantities.

Example 1.2: N distinguishable independent particles

Consider a system of N distinguishable independent particles each of which can exist in one of two states separated by an energy ϵ. Find the total energy of the system.

Thermodynamics

Thermodynamics is a method for discussing the macroscopic behavior of physical system phenomenologically. The first law of thermodynamics states that

In thermodynamics, one basic assumption is that, for any system, there exist a state function that has a unique value for given thermal conditions, e.g. $S = S(E, V, N)$. The internal energy E of a system is a function of state, which means that a system undergoes the same change in E when we move it from one equilibrium state to another, irrespective of which route we take through parameter space.

Consider the differential form of $S = S(E, V, N)$:

Comparing with

The Helmholtz free energy, F, is related to the energy E by a Legendre transformation:

This implies that the natural variables for F are T, V, and N, and we can therefore write $F = F(T, V, N)$. Legendre transformations allow one to change from one type of energy or free energy to another, by changing from one set of independent variables (e.g. E, V, and N) to another (T, V, and N).

Thermodynamic variables for a magnet

First law: $\mathrm{d}U = T\mathrm{d}S - M\mathrm{d}H$

Partition function

$$\mathcal{Z}(T,H) = \sum_r e^{-\beta E_r}$$

$$\downarrow$$

Free energy

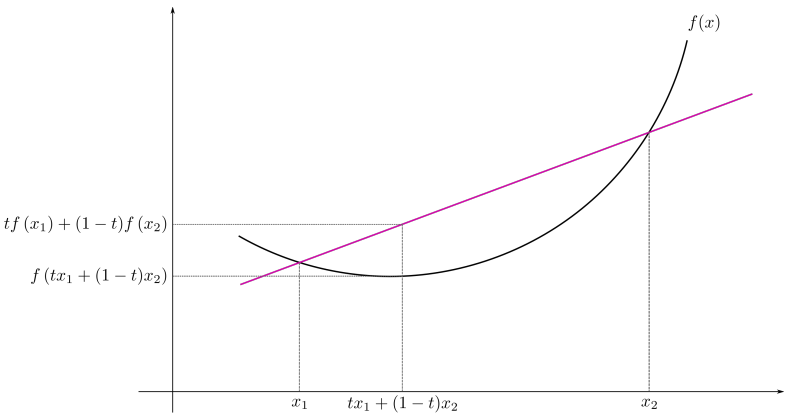
$$\mathcal{F} = -kT \ln \mathcal{Z}$$

Internal energy	*Entropy*	*Magnetization*
$U = -\frac{\partial \ln \mathcal{Z}}{\partial \beta}$	$S = -\left(\frac{\partial \mathcal{F}}{\partial T}\right)_H = (U - \mathcal{F})/T$	$M = -\left(\frac{\partial \mathcal{F}}{\partial H}\right)_T$
$\downarrow$	$\downarrow$	$\downarrow$
Specific heat (constant H)	*Specific heat (constant $X = H, M$)*	*Isothermal susceptibility*
$C_H = \left(\frac{\partial U}{\partial T}\right)_H$	$C_X = T\left(\frac{\partial S}{\partial T}\right)_X$	$\chi_T = \left(\frac{\partial M}{\partial H}\right)_T$

Convex/concave function

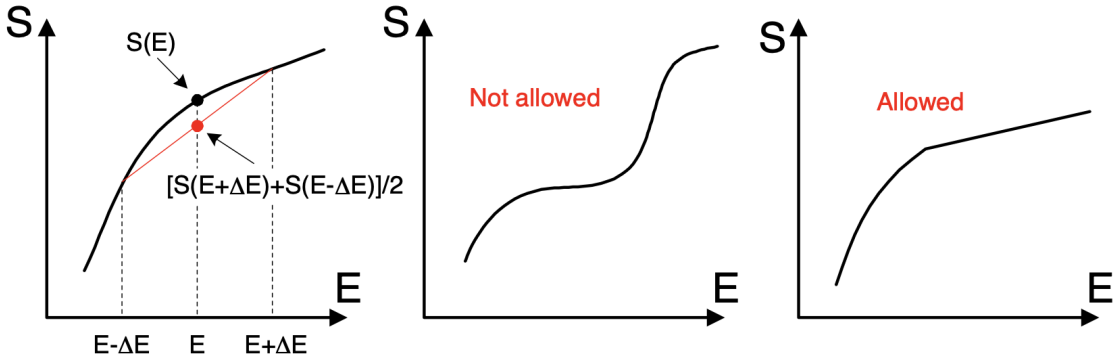
A function $f(x)$ is called a convex function of x if:

for all x_1 and x_2.



Alternatively, a convex function of a single variable $f(x)$ is one for which $f''(x) > 0$ everywhere. If the equality sign is reversed, then function is said to be concave.

The entropy S **increases** in the relaxation of a prepared macroscopic fluctuation to equilibrium when the appropriate natural variables are held fixed, eventually reaches a **maximum** in equilibrium (why?).



Interchanging the E and S axes, we see that the $E(S)$ curve must be **convex** i.e. any chord must lie above the curve.

On the other hand, thermodynamic potential (e.g F, G, etc) **decreases** in the relaxation of a prepared macroscopic fluctuation to equilibrium when the appropriate natural variables are held fixed, and attains a minimum in equilibrium.

A **Legendre transformation** to an intensive conjugate variable T, μ, P **reverses** the convexity. Thus the thermodynamic potentials F, G, ... are **concave** functions of the dependence on their intensive natural variables T, μ, P. e.g. Helmholtz free energy $F(T, N, V)$ is a concave function of T, and a convex function of N and V; Gibbs free energy $G(T, P, N)$ is a concave function of T and P, and a convex function of N.

Example 1.3: Concavity of F w.r.t. T

Show that F is a concave function of T.

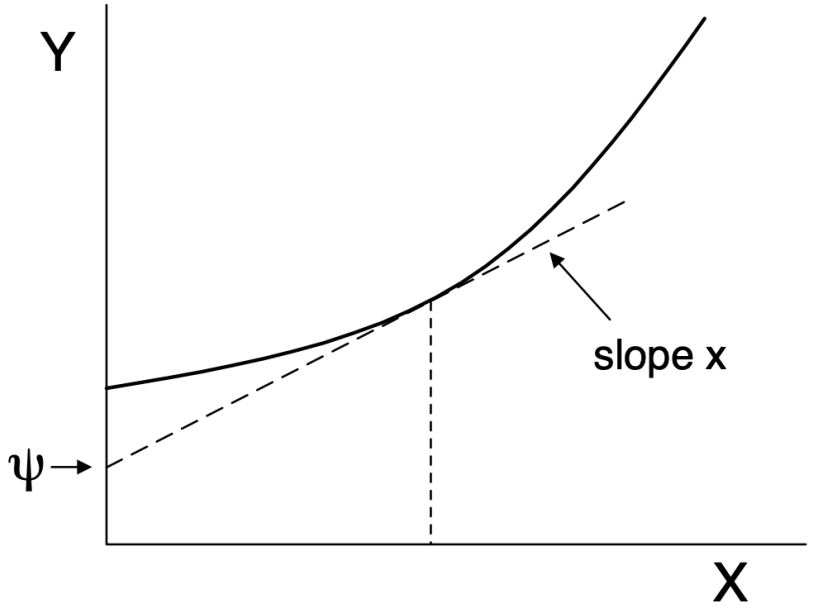
Indeed, we see that

where C_H is the specific heat at constant field. Additionally, one can show that

Since the second derivative of F with respect to T and H are negative, it is a concave function of both variables.

Mathematics of Legendre Transformation

Since we can consider each pair of variables separately (i.e. S, T or P, V etc.), lets look at a thermodynamic potential $Y(X)$ with thermodynamic identity $dY = xdX$ introducing the variable x conjugate to (paired with) X. Let's suppose that X is extensive, and then x is intensive.



Geometrical Picture of a Legendre transformation $Y(X) \to \psi(x)$.

Suppose we want to go from a representation in terms of X to one in terms of the intensive variable x defined by

We use the idea of "Pluecker line geometry" that tells us that the curve $Y(X)$ may be specified in terms of the tangents to the curve at each point: *the curve is the envelope of the tangents*. The tangent at the point X can be specified by the slope $x(X)$ and the intercept ψ on the Y axis which by simple geometry is

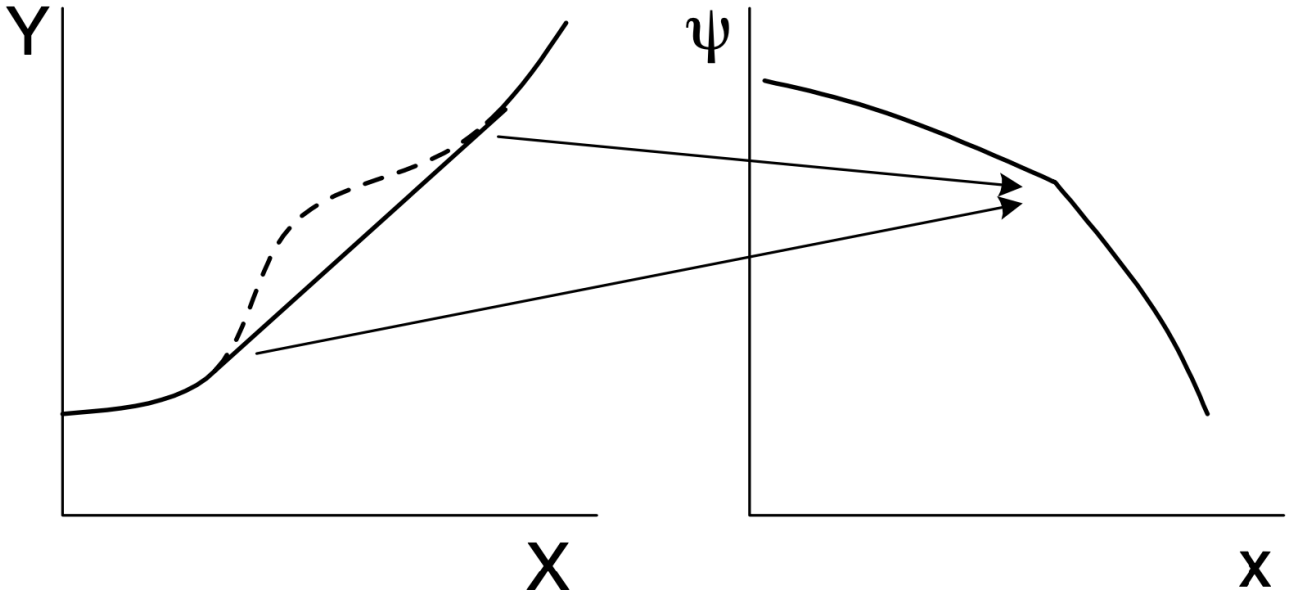
This is a Legendre transformation: $Y(X) \to \psi(x)$.

Furthermore we have

and then

so that if $Y(X)$ is smooth and **convex** $(\partial^2 Y/\partial X^2 > 0)$ then $\psi(x)$ will be smooth and **concave**.

We see that the Legendre transformation of a $Y(X)$ is formed by the "common tangent" construction. (Note: the dashed portion corresponds to unphysical results). The straight portion of $Y(X)$ translates into a single point of $\psi(x)$ where the slope changes discontinuously. Points (or lines) where the second derivative is zero or infinity are **critical points** and correspond to **phase transitions**.



Example 1.4: Molecular motors and free energies

Sethna 6.4 - Molecular motors and free energies.

Let's say the universe, which we'll consider from a microcanonical point of view, is composed of our system and the bath it sits in. The system can exchange heat, volume and distance with the external bath, so the thermodynamic identity reads:

$$dE_s = TdS_s - pdV_s + Fdx$$

where F is the force exerted on the system by the bath. Clearly, Fdx is thus the work done on the system by the bath for a displacement dx of the system. At equilibrium, the total entropy of the universe must be maximized. It is given by:

$$S_T \approx S_S + S_R(E - E_s, V - V_s, L - x)$$

Use the triple product relation:

$$\left.\frac{\partial S}{\partial x}\right|_E \left.\frac{\partial x}{\partial E}\right|_S \left.\frac{\partial E}{\partial S}\right|_x = -1$$

We find that

$$\left.\frac{\partial S}{\partial x}\right|_E = \frac{-F}{T}$$

For the entropy of the universe to be maximized, we see that $G_s - Fx$ must be minimized. Hence, if we start from a non equilibrium situation, we see that $G_s - Fx$ will decrease with time as the universe equilibrates.

Extra Notes